

Bellcourt Health Administrators: prior authorization under pressure

Case documentation. FDE hackathon, US healthcare. Lohith Devaramane, 6 October 2026.

In one paragraph. Bellcourt is late because requests arrive incomplete, and wrong because reviewers cannot find the rule that governs. It is not mainly a staffing problem. I built a Review Copilot (Agentic RAG) that selects the governing rules by client, service and date, checks each criterion with quoted evidence, and drafts the fax-back for missing information at intake. On the 120 cases Bellcourt's own QA team audited, it agrees with the auditor on **97.5%**. The original human decisions agreed on **56.7%**. It issued zero wrongful approvals and cannot deny.

1. The business and the prior authorization process

Bellcourt is a third-party administrator (TPA). It runs health plans for **38 self-funded employers** (about 453,000 people) who pay their staff's medical bills from their own money and pay Bellcourt a fee per employee per month. Since 2023 it also decides prior authorization requests on behalf of **Riverbend Health Plan** (about 153,700 members), which stays accountable to regulators and so audits Bellcourt and charges service credits when standards are missed. Prior authorization is a check before an expensive service: is it covered under this member's plan, and is it medically necessary under written criteria? Bellcourt handles about 310,000 requests a year.

Step	Today	Where it breaks (from the data pack)
1 Intake	46% of requests arrive by fax. A coordinator retypes 14 fields into PACE.	Faxes wait 16.9 hours on average before keying. PACE starts its clock at keying; the contract clock started at receipt.
2 Completeness	Incomplete requests are pended and the provider is contacted.	22.7% arrive incomplete (fax 32.3%). The wait for missing information averages about 3.5 days.
3 Nurse review	Nurse works out which rules apply, reads the record, approves or refers. Nurses cannot deny.	14.2 of 38 minutes per case go on finding the rules. PACE criteria screens are months out of date.
4 Physician review	A physician decides proposed denials.	Only 2 of 6 medical directors hold Arizona licenses; Arizona is 66% of Riverbend MA requests.
5 and 6 Letters, status, appeals	Reasons copied by hand into Word templates. Providers phone for status.	57% of calls are status checks. More than half of appeals are overturned.

Money: revenue grew 9.3% in three years (to \$112.4M) while clinical staff cost grew 27.7% and intake and call-center cost 23.7%. Operating margin fell from 14.0% to a forecast 8.0%. Riverbend service credits are \$1.9M this year, about a fifth of forecast profit. Riverbend opened a Corrective Action Plan in July 2026 and its contract ends on 31 December 2026.

2. Diagnosis, backed by the data pack

Leadership disagrees about the cause. I tested each claim against the files.

Claim	What the data shows	Verdict
VP Operations: it is a staffing problem.	Nurse FTE fell from 57.7 to 50.6, but on-time rate does not move with it (weekly correlation -0.05). Riverbend MA standard decisions were 87.0% on time in Q1 2025 at 98% of budgeted staff. All 64 late MA standard decisions arrived incomplete.	Mostly wrong
VP Operations: overturns come from new information at appeal.	Requests that arrived complete are overturned more often (56.7%) than incomplete ones (48.7%). Denials citing a retired policy version are overturned 91% of the time (30 of 33).	Not supported
Riverbend: it is a wrong-decision problem.	QA audit: 68 of 120 decisions correct (56.7%) against a 95% standard. 39 of 52 errors are wrong-rule errors: a memo overriding policy (18), a retired version (14), a client plan rule missed (7).	Supported
Senior nurse: the hard part is which rules apply.	34% of 2026 denials (94 of 274) cite a policy version already retired on the date of service. 72 lumbar MRI requests were denied under a void memo; 18 of 21 appeals were overturned.	Supported
Intake: a third of faxes are missing something.	66 of 69 late decisions arrived incomplete. Requests that arrive complete are on time 99.8% of the time. Late cases spend 122 of 199 hours waiting for missing information.	Supported

Sources: pa_requests_2025_2026.csv, qa_audit_sample_2026.csv, nurse_staffing_weekly.csv, call_center_log, GOV-01, UM-MEMO-2025-19, process documents. Caveat: the pack does not say the QA sample is random, so 56.7% is not extrapolated to all decisions; the request file independently shows the retired-version problem.

3. Priorities and business impact

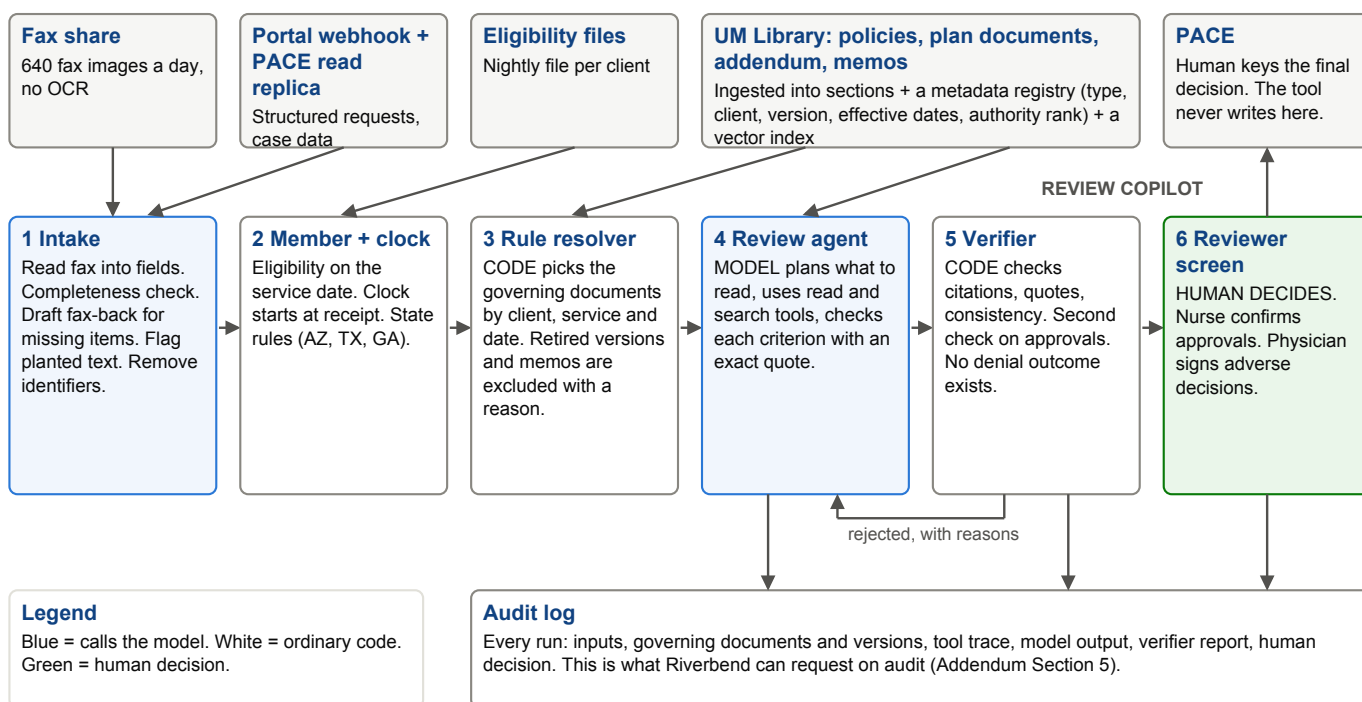
Rank	Problem	Evidence	Impact	Fix and pattern
1	Wrong rule applied. Reviewers cannot find which plan document, policy version or Medicare rule governs.	QA accuracy 56.7% vs 95%. MA overturn rate 56.2% vs a 30% trigger. 14.2 of 38 minutes spent hunting rules.	Contract risk with Riverbend and CMS audit exposure. Rule-hunting equals 18.9 of 50.6 nurse FTE.	Governing-rule copilot. Agentic RAG. Built.
2	Clock lost at intake. Fax, incomplete, days of waiting.	66 of 69 late decisions arrived incomplete. MA timeliness 90.1% vs 97%.	Direct cause of the \$1.9M credits and the Corrective Action Plan.	Completeness check and fax-back draft at receipt. AI agent step. Built as a slice.
3	No visibility. Providers phone for status; employers get a quarterly PDF.	67% of calls ask where a request is. 20% abandoned.	Call-center cost, Harlan renewal (\$8.1M a year).	Status lookup, requirements assistant (plain RAG). Later.

Problem 1 is first because it is the only fix that makes Bellcourt both faster and more defensible, which is the CEO's brief; it is the cause Riverbend named; and its rule engine is the foundation for the other two. Problem 2 drives the credits, so the build includes its highest-value slice: catching incompleteness in the first minute instead of after 17 hours.

4. The solution and where it fits

Bellcourt Review Copilot runs beside PACE and never writes to it. For each request it (1) reads the request, including un-keyed fax images; (2) checks completeness and drafts a fax-back listing exactly what is missing; (3) confirms eligibility and starts the clock from receipt; (4) selects the governing documents in code; (5) lets the model plan what to read and check each criterion with an exact quote; (6) verifies the answer; (7) hands a recommendation to a person. The working prototype covers the whole path: a case is raised by form, by uploading a case file, or by uploading a fax image or PDF; it joins a priority queue (overdue, urgent, due soon, standard); a nurse or physician records the decision; and a decision letter with the specific reason, the provision relied on and appeal rights is created, downloadable as PDF and sent by email. Sign-in is required and roles are enforced on the server.

BELLCOURT SYSTEMS (READ-ONLY). PACE STAYS THE SYSTEM OF RECORD.



Architecture. Blue boxes call the model, white boxes are code, green is the human decision. Full-page version: docs/Architecture_Diagram.pdf.

Design decision	Why
Rule selection is code, not the model	The failure at Bellcourt is choosing the wrong version. Versions 1 and 2 of a policy are near-identical text and the void memo looks highly relevant, so similarity search alone would repeat the error. A registry with effective dates and authority rank (GOV-01) decides; the model can only read documents that are in force.
Agentic, but bounded	The model plans its own retrieval and follows cross-references (plan section to amendment, policy to MP-117). The loop is capped, a safety net always reads exclusions and amendments, and a verifier rejects bad citations, invented quotes and inconsistent outcomes.

Design decision	Why
Four outcomes, no denial	Approve-ready, need information, route to physician, route as not covered. Required by the Riverbend addendum and state law. A second, skeptical model pass checks every approval because a wrong approval is the error no human would catch.
Provider text is untrusted	Fax and note text is data. Planted instructions are flagged, cannot be used as evidence, and do not change the outcome.

5. Evidence it works

Test	Data	Result
Agreement with the QA auditor	120 audited cases with the auditor's correct decision	117 of 120 (97.5%) . Original human decisions: 68 of 120 (56.7%)
Cases humans got wrong for a wrong-rule reason	Memo conflict, retired version, client rule missed	18 of 18, 14 of 14, 6 of 7
Governing document and version	Same 120 cases	118 of 120 correct
Wrongful approvals / denials issued	Same 120 cases	0 wrongful approvals. 0 denials (not possible by design)
Open queue	30 live cases incl. 13 fax images, against my answer key	29 of 30. Misses fall on the cautious side
Adversarial, incomplete, wrong inputs	Planted instructions (10), policy time-travel (7), missing data and bad inputs (7)	23 of 24 pass. The one failure ended in manual review, not a wrong answer
Deterministic unit tests	Resolver, clock, eligibility, de-identification, verifier, login and roles	36 of 36 pass

Model: google/gemini-2.5-flash through OpenRouter, about 6.5 seconds and a third of a US cent per case. Limits: results vary slightly between runs (27 to 29 of 30 on the open queue across runs); prompts were tuned while looking at these cases, so a fresh audit sample is the right next test; only 6 of 38 employer plan documents were in the pack. Full tables: docs/EVIDENCE.md and the Evidence tab of the app.

6. Success metrics

Measure	Today	Target	How measured
Riverbend audit accuracy	72% on MA cases in the QA sample	At least 95%	Monthly QA audit
MA appeal overturn rate	56.2%	Under 30%	Appeals file
MA standard decisions on time	90.1% (Q2 2026)	At least 97%	Clock from receipt, not keying
Denials citing a retired policy version	34% of 2026 denials	Zero	Audit log
Nurse minutes per case	38.0, of which 14.2 finding rules	About 25	Repeat the time study
Hours from fax receipt to missing-information request	About 17 before keying	Under 1	Audit log timestamps
Adverse decisions made by the tool	n/a	Zero, always	Audit log and tests

7. Compliance and risk

Rule	Source	How the design meets it
No tool may deny, delay or modify a request; a qualified human makes every adverse decision	Riverbend addendum s5; CEO brief	No denial outcome exists. The fax-back is a draft a person sends. Tests assert zero denials.
60 days' written notice before any AI tool is used	Riverbend addendum s5	Send notice on day 0. Pilot on self-funded plans (ERISA; state AI laws do not bind them) while the notice runs.
Produce inputs and outputs for any case on audit	Riverbend addendum s5	Per-run audit record: inputs, sources and versions, tool trace, output, verifier report, human decision.
Arizona: licensed medical director signs denials. Texas: disclose AI use. Georgia: human review before denial	Addendum s4; REG-02	State flags route Arizona cases to a licensed director and add the Texas disclosure to the letter draft.
Patient data is PHI	IT interview; HIPAA	Name, DOB, phone, record number and member ID are removed before the review step. Production runs inside the Azure tenant under the existing agreement.
Provider content is outside Bellcourt's control	Pack README; intake interview	Treated as data; planted instructions flagged and ignored (10 of 10 in tests, plus faxes 8106 and 8120).

Main risks: model error (mitigated by code-selected rules, verifier, second check, human decision); stale registry (mitigated by making registry updates part of the Clinical Policy Committee release and the plan-amendment process); over-reliance (mitigated by showing evidence quotes and the excluded documents, and by monthly QA on tool-assisted cases).